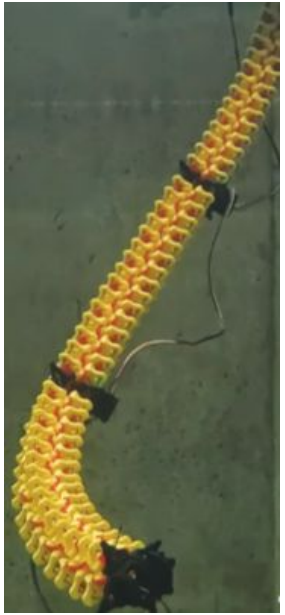
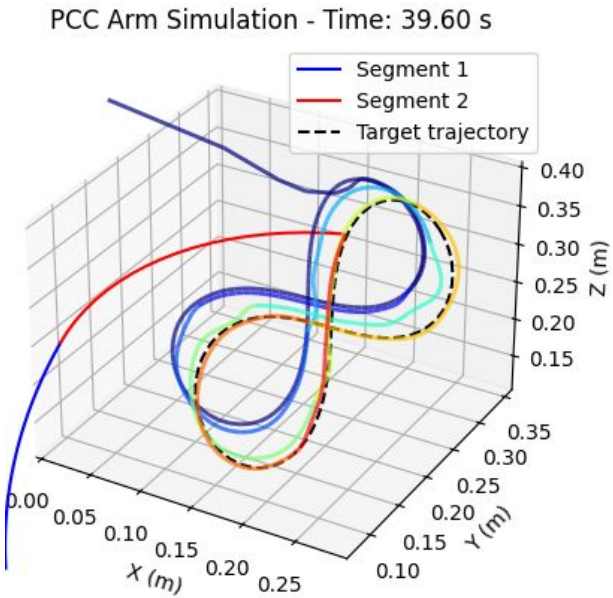
Adaptive Model Predictive Control for Compliant Manipulators





Z. Chen, Q. Guan, J. Hughes, A. Menciassi and C. Stefanini, "A Versatile Neural Network Configuration Space Planning and Control Strategy for Modular Soft Robot Arms,"
Walker, Kyle L., et al. “Closed-Loop Control and Disturbance Mitigation of an Underwater Multi-Segment Continuum Manipulator.”

This work

- Robots or controllers are often designed for **one** environment only.
- The goal is to develop a **data-driven**, **MPC** framework for compliant manipulators (non-linear) able to modify the system dynamics online when **changing environment** for example water/air or hysteresis

- Idea: keep it simple to remain fast:

  For each segment, the parameters are $p_i = (m_i, \beta_i, k_{\phi_i}, k_{\theta_i})$

We add adaptive parameter on each of them:

$$p_i^{\text{tot}} = (m_i + m_{i,a}, \beta_i + \beta_{i,a}, k_{\phi_i} + k_{\phi_i,a}, k_{\theta_i} + k_{\theta_i,a})$$

This work

Updated by minimizing the predicted states and measured states over the last x steps:

Algorithm 1 Adaptive NMPC loop

for $k = 0, 1, \ldots, T_{\text{sim}}/\Delta t - 1$ **do**

  (Reference assembly) Build $X_{k:k+N}^{\text{task space, ref}} \leftarrow [x_k^{\text{task space, ref}}, \ldots, x_{k+N}^{\text{task space, ref}}]$

  (NMPC solve) $\tau_{\text{tendon, k}} \leftarrow \text{acados}(\hat{x}_k, X_{k:k+N}^{\text{task space, ref}}, p_k)$

  (Apply to true system) $x_{k+1}^{\text{true}} \leftarrow f_{\text{true}}(x_k^{\text{true}}, \tau_{\text{tendon, k}})$

  (Measurement update) $\hat{x}_{k+1} \leftarrow x_{k+1}^{\text{true}} + \varepsilon_{k+1}$ ▷ add measurement noise

  (Parameter adaptation) $p_{k+1} \leftarrow p_k$

  Solve Ipopt update:

  $$p_{k+1} = \underset{p}{\text{argmin}} \sum_{i=k-N_{\text{prev}}}^{k-1} \|x_{i+1}^{meas} - F(x_i^{meas}, \tau_{\text{tendon, i}}; p)\|_2^2 + \lambda \|p - p_{\text{prev}}\|^2$$

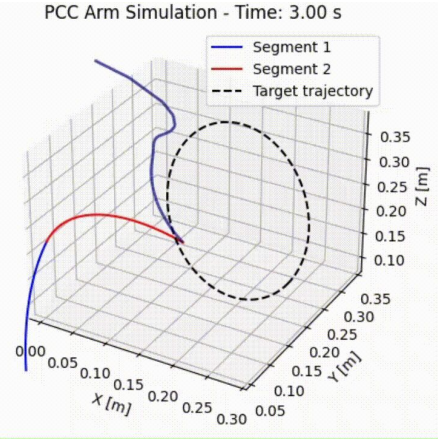
end for

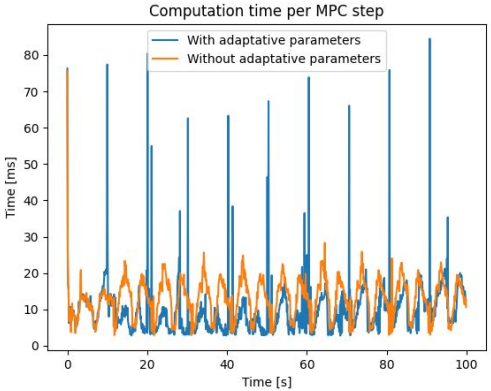
Simulation Setup - Overview

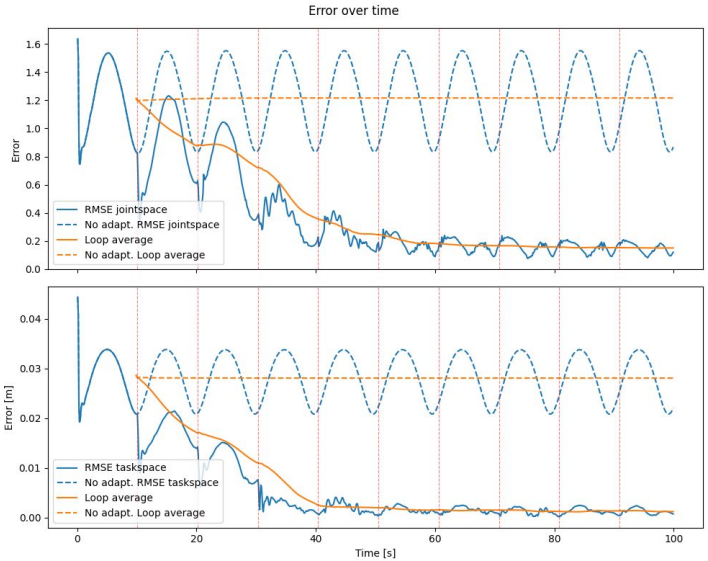
We have a simulation of a n-segment arm with:

- Taskspace trajectory following (circles, rectangles, lemniscates)
- Affected by fluid through buoyancy, added mass and drag
- Real time solving (>10Hz with adaptation, >50Hz without)
- Control on tendon tension directly (for a direct mapping to current control)

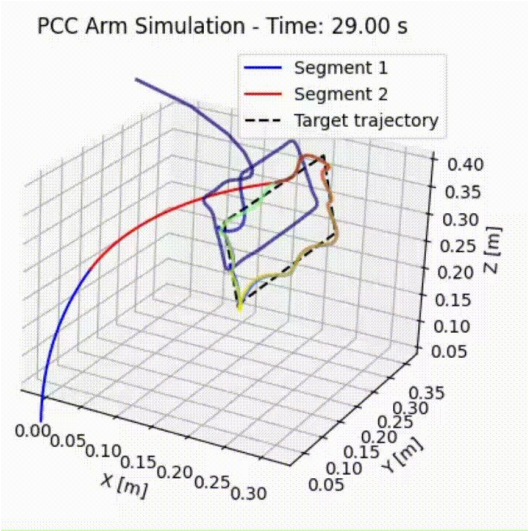
Simulation 2-segments

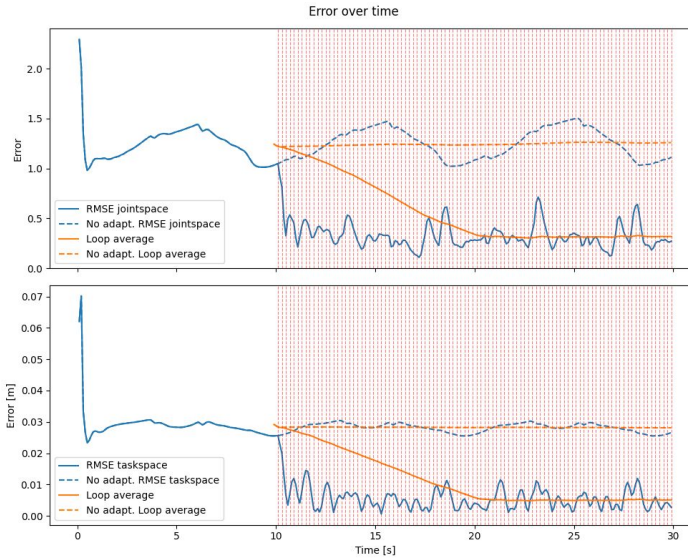


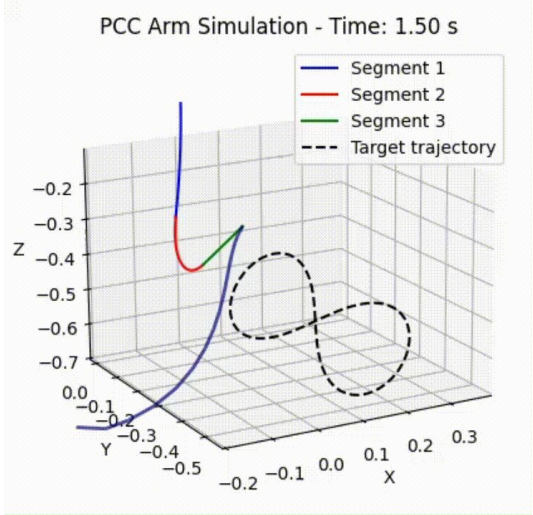




Other simulations:



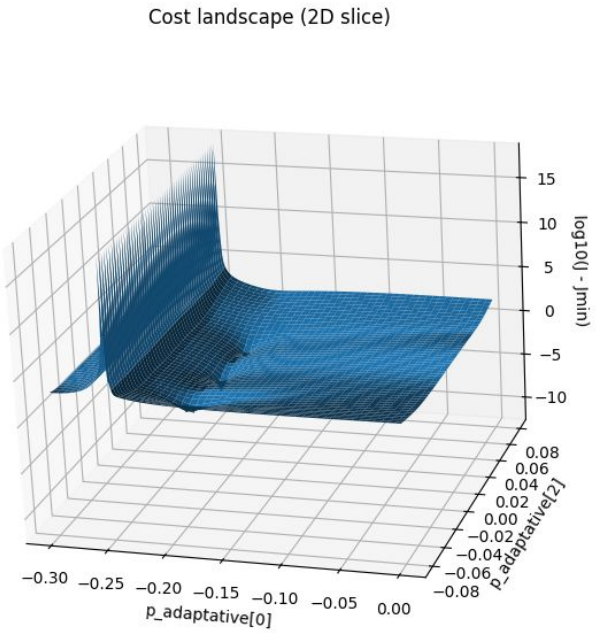




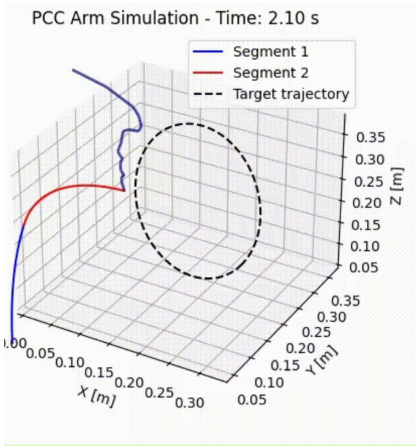
Notes:

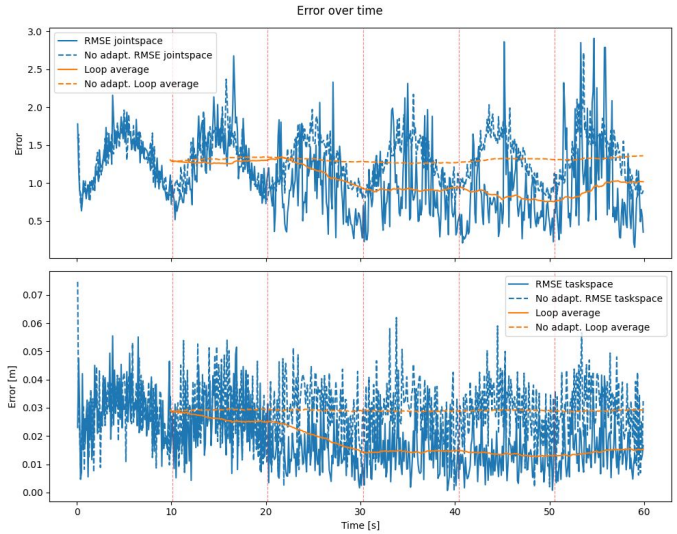
Non-convex

Can become unstable if not careful



Not very robust to noise





Real arm Setup - Overview

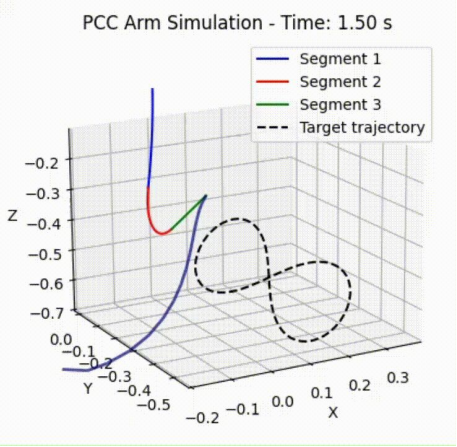
Tried to implement on the real arm:

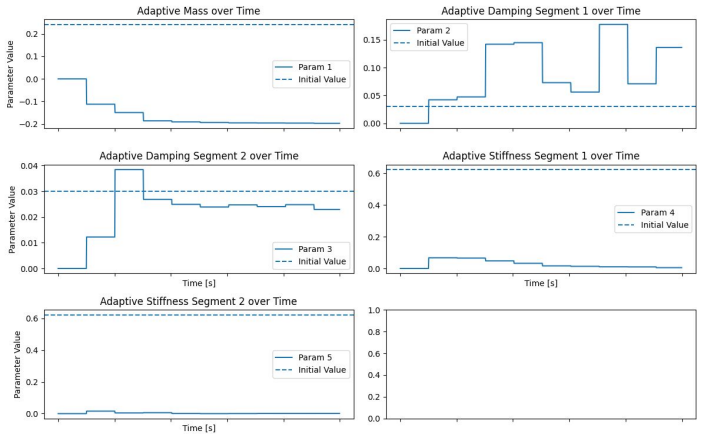
- Some mechanical issues at first
- Cyclic inputs does not make cyclic movement

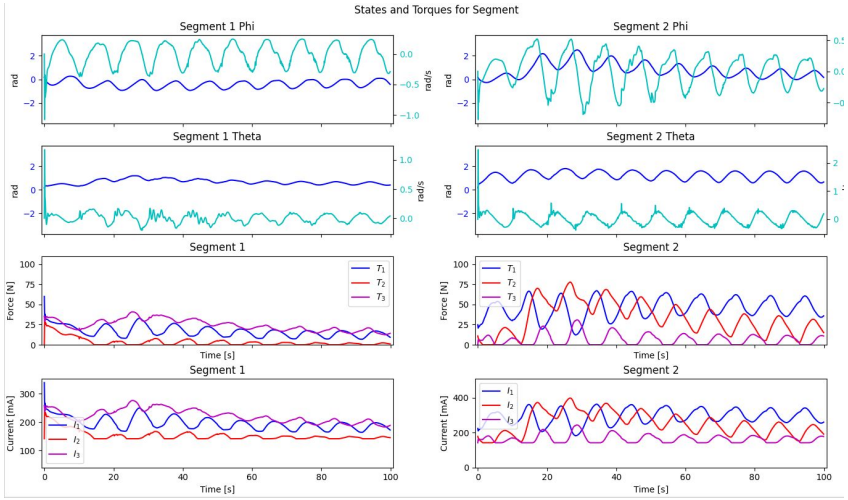
Conclusion

- This idea works in simulation under modeled mismatch and has the advantage of **not requiring any previous data** collection and easily integratable. However, testing on a real arm needs to be performed.

Any Questions?




Adaptive Mass over Time
Parameter Value
Param 1
Initial Value
Adaptive Damping Segment 1 over Time
Param 2
Initial Value
Adaptive Damping Segment 2 over Time
Parameter Value
Time [s]
Param 3
Initial Value
Adaptive Stiffness Segment 1 over Time
Param 4
Initial Value
Time [s]
Adaptive Stiffness Segment 2 over Time
Parameter Value
Time [s]
Param 5
Initial Value


States and Torques for Segment
Segment 1 Phi
Segment 2 Phi
rad
rad/s
Segment 1 Theta
Segment 2 Theta
Segment 1
Segment 2
Force [N]
Time [s]
Current [mA]
Time [s]